Guillaume Koenig

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EDUCATION

University of Toronto, non-degree student (2025 to present)

Cumulative GPA 4.00. BIO120 (A+), BIO130 (A+), STA237 (A+), STA238 (A+), STA302 (A). In progress: STA314, MAT224.

Illinois Institute of Technology (dual degree, USA), Master of Computer Science (2010)

ENSEIRB-MATMECA (dual degree, France), M. Eng. in Computer Science (2011)

RESEARCH

CSB498 Independent Research, Cell & Systems Biology, U of T (May to Aug 2026, A+)

Supervisor: Prof. Alan Moses | Outcome: moses-lab.github.io/faidr-fit

Built FAIDR-Fit, a web tool that lets researchers define a group of related proteins and see which conserved molecular features of their intrinsically disordered regions (IDRs) are associated with that group. It uses L1-penalized logistic regression over 19,302 human IDRs. Specified and shipped the model in JavaScript so it runs entirely in the browser (about 2 seconds, no backend). Validated against R's glmnet on a known protein group, matching feature strength and recall. Applied it to a new splicing protein set, where it flagged additional proteins with known splicing role. Used AI to generate code from my specifications, and added tests to check against R's glmnet reference outputs.

EXPERIENCE

Amazon (2019 to 2025)

Worked as software engineer in different organizations within Amazon.

- Redis is the in-memory cache behind Amazon ElastiCache and MemoryDB (1M+ managed instances). Led the design of an automated testing framework validating linearizability of workloads for durable Redis, running on every commit. Built a workload fuzzer from command syntax specs that independently rediscovered a known upstream crash (CVE-2023-28425) and found several bugs in an unreleased feature before launch. Contributed a radix tree bugfix to upstream Redis.
- Also at Amazon: In supply chain, launched a compliance and shipment portal for China-to-US imports; built infrastructure tooling to support Amazon's voice assistant Alexa.

Scality (2018 to 2019)

- Scality provides software for on-premise object storage, similar to storing in the cloud (S3). Worked on a filesystem in userland, building on top of the object storage primitive.

Leboncoin.fr (2013 to 2017)

- Leboncoin is France's largest second-hand e-commerce website, with 28M monthly active users. Designed and implemented traffic distribution for content moderation, with site receiving ~800K new ads/day. Debugged race conditions to improve system stability.

Philog (2011 to 2013)

- Software company specializing in embedded software. Dealt with Linux kernel driver integrations.

SKILLS

Statistics: R, glmnet, L1-penalized regression

Domains: distributed systems, in-memory databases, system programming

Tools: C, Rust, Python, Linux